Fiscal Year:	FY 2026	**Task Last Updated:**	FY 07/03/2025
PI Name:	Laurie, Steven Ph.D.		
Project Title:	Spaceflight Associated Neuro-Ocular Syndrome Directed Science Objective (SANS DSO)		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	steven.laurie@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-244-0029
Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Cardiovascular and Vision Laboratory		
PI Address 2:	2400 NASA Pkwy		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-2749	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2024	**End Date:**	07/03/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 07/03/2025 per C. Ribeiro/HHC (Ed., 10/24/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brunstetter, Tyson O.D., Ph.D. (NASA Johnson Space Center) Huang, Alex M.D., Ph.D. (University of California, San Diego) Kramer, Larry M.D. (University of Texas Health Science Center at Houston) Macias, Brandon Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The goal of the SANS DSO project is to collect key spaceflight associated neuro-ocular syndrome (SANS) outcome measures in crewmembers who develop SANS findings and are not already enrolled in the "Complement of Integrated Protocols for Human Exploration Research: Investigating Structure and Function of the Eye" (CIPHER: ISAFE) research study. Key data such as optical coherence tomography (OCT) images, vascular ultrasound, brain magnetic resonance imaging (MRI), and vision tests will be shared from MedB data collections. Additional research measures will be collected by the investigator team, including electroretinography (ERG), intravenous fluorescein angiography (IVFA), and OCT angiography. These data will allow SANS stakeholders to better characterize SANS findings and investigate relationships between datasets, especially for those presenting with the most significant SANS findings.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research.
Research Impact/Earth Benefits:	
Task Progress:	Per NASA Johnson Space Center (JSC), this investigation has been discontinued. No further reporting is expected (Ed., 7/3/25).
Bibliography Type:	Description: (Last Updated: 12/10/2025)